Assignment 4: Implementing Control Structures

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MSCS 632 - Advanced Programming Languages

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Implemented in Python and JavaScript. Each program schedules 12 employees over 7 days and 3 shifts with at least 2 employees per shift, at most one shift per day, and at most 5 days per week per employee, and supports ranked shift preferences.

GitHub repository: https://github.com/anphan7/MSCS632-Assignment-4

Python output

```
$ python3 scheduler.py --demo

================================================================================================================
                                              WEEKLY SCHEDULE
================================================================================================================
           Morning                          Afternoon                        Evening
----------------------------------------------------------------------------------------------------------------
Monday     Bob, Carol, Niaj, Heidi          Judy, Dave, Frank, Mallory       Erin, Ivan, Alice, Grace
Tuesday    Ivan, Carol, Niaj, Bob           Judy, Alice, Grace, Dave         Frank, Mallory, Heidi, Erin
Wednesday  Heidi, Carol, Niaj, Mallory      Alice, Dave, Judy, Grace         Frank, Erin, Bob, Ivan
Thursday   Erin, Heidi                      Dave, Mallory                    Grace, Frank
Friday     Bob, Niaj                        Judy, Alice                      Ivan, Carol
Saturday   Niaj, Heidi                      Carol, Dave                      Ivan, Bob
Sunday     Erin, Alice                      Mallory, Grace                   Frank, Judy
================================================================================================================
```

Command: python3 scheduler.py --demo

JavaScript output

```
$ node scheduler.js --demo

================================================================================================================
                                            WEEKLY SCHEDULE
================================================================================================================
            Morning                         Afternoon                       Evening
----------------------------------------------------------------------------------------------------------------
Monday      Ivan, Mallory, Niaj, Frank      Carol, Erin, Heidi, Bob         Dave, Alice, Grace, Judy
Tuesday     Niaj, Mallory, Ivan, Dave       Carol, Bob, Heidi, Erin         Grace, Frank, Judy, Alice
Wednesday   Mallory, Ivan, Niaj, Erin       Carol, Bob, Heidi, Judy         Frank, Alice, Dave, Grace
Thursday    Mallory, Alice                  Heidi, Erin                     Dave, Judy
Friday      Ivan, Niaj                      Carol, Bob                      Grace, Frank
Saturday    Ivan, Judy                      Bob, Erin                       Grace, Frank
Sunday      Mallory, Niaj                   Carol, Heidi                    Alice, Dave
================================================================================================================
```

Command: node scheduler.js --demo